

Name: _____

Maths Class Teacher: _____

SYDNEY TECHNICAL HIGH SCHOOL



Extension 1 Mathematics

HSC Assessment Task 1

Dec 2010

General Instructions

- Working time – 70 minutes
- Write using **black or blue pen**
- Board-approved calculators may be used
- **All necessary working** should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Start each question on a **new page**.
- Place your papers **in order** with the question paper on top and staple or pin them.

Total Marks - 50

- Attempt Questions 1 – 6
- Mark values are shown with the questions.

(For markers use only)

Q1	Q2	Q3	Q4	Q5	Q6	Total
10	8	8	8	8	8	50

Question 1**Marks**

- a) The sum of the first two terms of a geometric progression is -4 and the sum of the fourth and fifth terms is 108 . Calculate:

- (i) the common ratio 2
- (ii) the seventh term. 2

- b) For the geometric progression $3, 9, 27, \dots$

- (i) Find the sum of n terms 1
- (ii) Determine how many terms must be taken for the sum to exceed 10000 . 2

- c) Prove that, if x is positive, the sum of 3

$$1 + \frac{x}{1+x} + \frac{x^2}{(1+x)^2} + \frac{x^3}{(1+x)^3} + \dots$$

never exceeds $1 + x$.

Question 2**Marks**

- a) AB is a diameter and AC is a chord of a circle whose centre is O .
 D is the midpoint of the arc BC .

- (i) Construct a diagram showing the above information. 1
- (ii) Prove that OD is parallel to AC . 3

- b) Prove by mathematical induction that 4

$$\sum_{r=1}^n r(r+1) = \frac{n(n+1)(n+2)}{3}$$

Question 3**Marks**

- a) Two circles intersect in X and Y and P is a point on one of them. PX and PY , when produced, meet the other circle in M and N respectively.
- (i) Construct a diagram showing all relevant information. **1**
- (ii) Prove that the tangent at P is parallel to MN . **3**
- b) Find, without deriving, the locus of a point $P(x,y)$ which moves so that its distance from the fixed point $(0,4)$ is always equal to its perpendicular distance from the fixed line $y = -4$. **1**
- c) Show that the equation of the chord joining the points where $x = x_1$ and $x = x_2$ on the parabola $x^2 = y$ is **3**
- $$y = xx_1 + xx_2 - x_1x_2.$$

Question 4**Marks**

A woman is considering borrowing \$24 000 to finance renovations to her house. The interest rate is 9% per annum compounded monthly on the balance owing.

- a) If A_n represents the amount owing after n months and, using M to represent the monthly repayment, write an expression to show the amount owing after 1 month. **1**
- b) Write another expression showing the amount owing after 2 months. **1**
- c) Construct an expression to express the amount owing after n months. **2**
- d) Calculate the monthly instalment (to the nearest dollar) if the loan is to be repaid in 8 years. **3**
- e) What is the full amount of interest paid (to the nearest dollar)? **1**

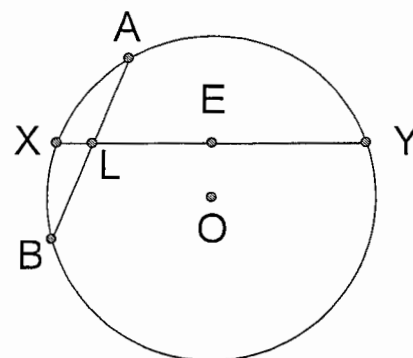
Question 5**Marks**

- a) AB and XY are chords in a circle with centre O . XY cuts AB in L , which is the midpoint of AB . E is the midpoint of XY .

3

Prove that XY is greater than AB .

[Hint: Construct OL and OE .]



- b) In a proof by mathematical induction, it is assumed that $8^k - 5^k$ is divisible by 3 for a positive integer value of k .

3

Using this assumption, show that this must also be true for $k+1$.

- c) A circle is drawn with one of the equal sides of an isosceles triangle as diameter.

2

Show that the circle passes through the midpoint of the base of the isosceles triangle.

Question 6**Marks**

- a) On the parabola $x^2 = 4ay$, the point P has coordinates $(2ap, ap^2)$.

2

Show that the locus of the midpoints of chords PO where O is the vertex is another parabola, $x^2 = 2ay$.

- b) Tangents are drawn to a parabola $x^2 = 4y$ from an external point $A(x_1, y_1)$, touching the parabola at P and Q .

(i) Write the equation of the chord of contact.

1

(ii) Prove that the midpoint, M , of PQ is the point $(x_1, \frac{1}{2}x_1^2 - y_1)$.

3

(iii) If A moves along the straight line $y = x - 1$, find the equation of the locus of M .

2

End of Exam

HSC Assessment One

Extension 1 2010

Q1 a) $T_1 + T_2 = -4$ — (i)
 $T_4 + T_5 = 108$ — (ii)

(i) From (i) $a + ar = -4$ — (iii)

From (ii) $ar^3 + ar^4 = 108$ — (iv)

$\therefore ar^3(1+r) = 108$ — (v)

from (iii) $a(1+r) = -4$

sub into (v) $ar^3 \times -4 = 108$

$\therefore r^3 = \frac{108}{-4}$
 $= -27$

(ii) $\therefore r = -3$

sub into (iii)

$a - 3a = -4$

$\therefore -2a = -4$

$\therefore a = 2$

$\therefore T_7 = 2 \times (-3)^6$
 $= 1458$

b) $3, 9, 27, \dots$
 $a=3, r=3$

(i) $S_n = \frac{a(r^n - 1)}{r - 1}$
 $= \frac{3(3^n - 1)}{2}$

Now $\frac{3(3^n - 1)}{2} > 10000$

$3(3^n - 1) > 20000$

$\therefore 3^n - 1 > \frac{20000}{3}$

$\therefore 3^{n+1} - 3 > 20000$

$\therefore 3^{n+1} > 20003$

$\therefore \log(3^{n+1}) > \log 20003$

$\therefore (n+1) \log 3 > \log 20003$

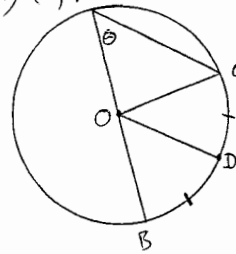
$\therefore n+1 > \frac{\log 20003}{\log 3}$

$\approx 9.01467..$

$\therefore n > 8.01467$

So $n=9$ for $S_n > 10000$

Q2 a) (i) A



(ii) Let $\angle BOC = \theta$

Now, $\angle CAB = 2\theta$ (Angle at centre is twice that at circumference)

$\angle COD = \angle BOD$ (angles on equal arcs)

$\therefore \angle BOD = \theta$

$\therefore AC \parallel OD$ (corresponding angles)

b) RTS $\sum_{i=1}^n r(n+i) = \frac{n(n+1)(n+2)}{3}$

i.e. $1.2 + 2.3 + \dots + n(n+1) = \frac{n(n+1)(n+2)}{3}$

(i) For $n=1$, LHS = 1.2
 $= 2$

RHS = $\frac{1 \times 2 \times 3}{3}$

$= 2$

$\therefore$ LHS = RHS
 $\therefore$ true for $n=1$.

(ii) Assume true for $n=k$

$\therefore 1.2 + 2.3 + \dots + k(k+1) = \frac{k(k+1)(k+2)}{3}$

For $n=k+1$,

RHS = $\frac{(k+1)(k+2)(k+3)}{3}$

LHS = $1.2 + 2.3 + \dots + k(k+1) + (k+1)(k+2)$

$= \frac{k(k+1)(k+2)}{3} + (k+1)(k+2)$

$= \frac{k(k+1)(k+2)}{3} + \frac{3(k+1)(k+2)}{3}$

$= \frac{(k+1)(k+2)(k+3)}{3}$

$=$ RHS

$\therefore$ if true for $n=k$, then true for $n=k+1$

(iii) If true for $n=k=1$, then true for $n=k+1=2$
 If true for $n=k=2$, then true for $n=k+1=3$
 etc.

$\therefore$ true for all positive integral values of n .

c) $1 + \frac{x}{1+x} + \frac{x^2}{(1+x)^2} + \dots$

GP where $a=1, r=\frac{x}{1+x}$

If $x>0$, then $0 < \frac{x}{1+x} < 1$

$\therefore |r| < 1$

$\therefore S_{\infty}$ exists.

$\therefore S_{\infty} = \frac{a}{1-r}$

$= \frac{1}{1 - \frac{x}{1+x}}$

$= \frac{1}{\frac{1+x-x}{1+x}}$

$= 1+x$

QED

b)

A hand-drawn graph of a parabola opening upwards with its vertex at the origin $(0,0)$. The y-axis is labeled y and the x-axis is labeled x . A point $(0, a)$ is marked on the y-axis. A point $P(x, y)$ is marked on the parabola. A horizontal line segment connects $(0, a)$ to $P(x, y)$, and a vertical line segment connects $P(x, y)$ to the x-axis. The label $y = -4$ is written below the x-axis.

c)

A Cartesian coordinate system showing the parabola $x^2 = y$. The vertex is at the origin. Two points on the curve are labeled (x_1, x_1^2) and (x_2, x_2^2) . A line segment connects these two points.

Let $\Theta = \hat{A}^T P \Pi$

Now, $\angle x = \theta$ (angle in alternate segment)

$\therefore \angle N = 180^\circ - \theta$ (supplementary angles)

$\therefore \angle PMN = \theta$ (opposite angles in cyclic quadrilateral are equal)
 $\therefore AB \parallel MN$ (alternate angles are equal)

Locus of P is a parabola, vertex $(0,0)$ with focal length, 4.

$$\therefore x^2 = 4 \times 4 \times y$$
$$\therefore x^2 = 16y$$

$$\frac{y - x_1^2}{x - x_1} = \frac{x_2^2 - x_1^2}{x_2 - x_1}$$

$$\therefore (y - x_1^2)(x_2 - x_1) = (x_2^2 - x_1^2)(x - x_1)$$

$$y x_2 - x_1 y - \cancel{x_1^2 x_2} + \cancel{x_1^3} = x_2^2 x - x_2^2 x_1 - \cancel{x_1^2 x} + \cancel{x_1^3}$$

$$\begin{aligned} \therefore y(x_2 - x_1) &= x_1^2 x_2 - x_2^2 x_1 + x x_2^2 - x x_1^2 \\ &= x_1 x_2 (x_1 - x_2) + x (x_2^2 - x_1^2) \\ &= x_1 x_2 (x_1 - x_2) + x (x_2 - x_1) (x_2 + x_1) \\ &= -x_1 x_2 (x_2 - x_1) + x (x_2 + x_1) (x_2 - x_1) \end{aligned}$$

$$\begin{aligned} y &= -x_1 x_2 + x(x_2 + x_1) \\ &= -x_1 x_2 + x x_2 + x x_1 \\ &= x x_1 + x x_2 - x_1 x_2 \end{aligned}$$

 $Q \leq D$
$$r = 0.09 \mu\text{m} = 0.0075 \mu\text{m}$$

a) $A_1 = 24000 \times 1.0075 - M$

$$b) A_2 = (24000 \times 1.0075 - 7) \times 1.0075 - 7$$
$$= 24000 \times 1.0075^2 - 7(1 + 1.0075)$$

$$c) A_n = 24000 \times 1.0075^n - 17 \left(1 + 1.0075 + 1.0075^2 + \dots + 1.0075^{n-1} \right)$$

$$d) A_{96} = 24000 \times 1.0075^{96} - M \left(1 \frac{(1.0075^{96} - 1)}{1.0075 - 1} \right)$$

$$= 24000 \times 1.0075^{96} - M \times 139.8562$$

$$\begin{aligned} \therefore 139.8562 M &= 24000 \times 1.0075^{46} \\ &= 49174.10948 \\ &\approx 351.60 \end{aligned}$$

$$\therefore M = \$352$$

$$\begin{aligned} \text{c) Interest} &= 352896 - 24000 \\ &= 33792 - 24000 \\ &= \$9792 \end{aligned}$$

